

Task 2: Multimodal Idiomaticity Representation

2.1 Exploratory Data Analysis

Three splits were used: 231 training, 27 validation, and 27 test samples, with columns — compound, sentence_type, sentence, image_name, image_caption, and label. There are 77 unique idiomatic compounds, each appearing exactly 3 times per split (1 sentence, 3 candidate images). Label 1 marks the correct image; label 0 marks wrong ones. No null values were found across any split. Sentence length averaged 22 words (std 5.9, range 11–37). The label split was 154 negatives to 77 positives — a 2:1 ratio that shaped the pos_weight used in all trained models.

2.2 Data Augmentation

Vamsi/T5_Paraphrase_Paws [4] was used to generate up to two paraphrases per training sentence (top-k=50, top-p=0.9, temperature=1.0). Outputs under 4 words or identical to the original were removed. This produced 459 new rows; the final training set held 690 samples (461 negative, 229 positive, pos_weight=2.01). All non-sentence fields — compound, image_name, image_caption, label — were carried over unchanged.

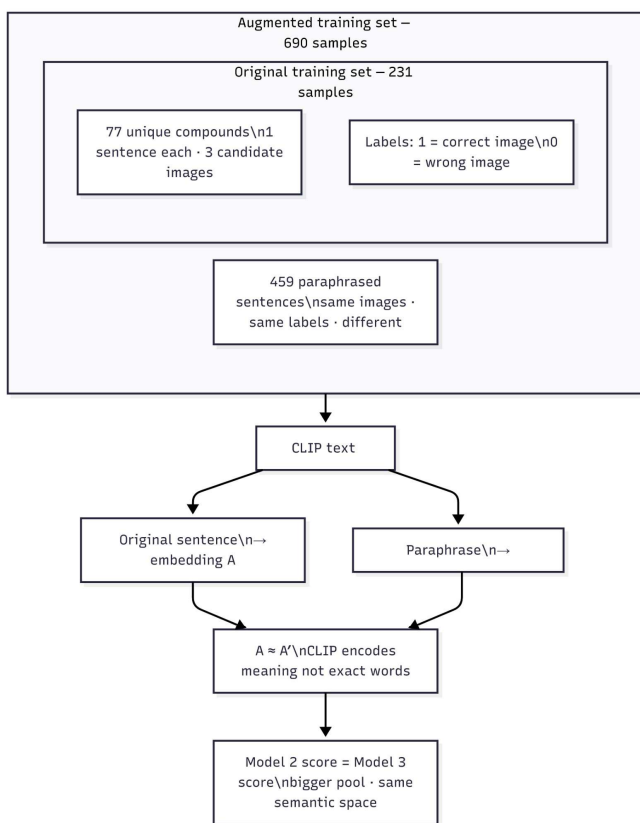


Fig. 4 — Augmentation structure: 231 → 690 via T5 paraphrase, and CLIP semantic equivalence rationale.

2.3 Pipeline

CLIP ViT-B/32 [5] encoded images to 512-d and text (compound + sentence) to 512-d. Both were L2-normalised then concatenated into a 1024-d vector — the shared input across all trained models. Feature extraction ran once and tensors were reused across all training runs.

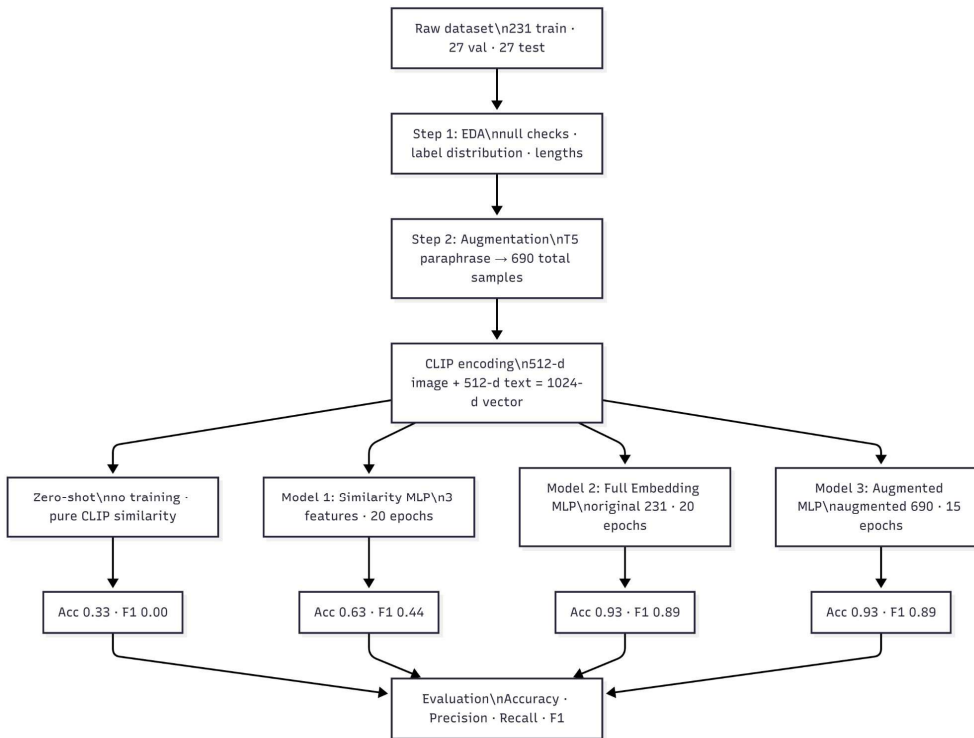


Fig. 5 — Full pipeline: EDA through augmentation, CLIP encoding, and all four prediction approaches.

2.4 Zero-Shot

Raw CLIP cosine similarity with no training. For each group of three candidates, the highest-scoring image gets prediction=1. Test: Accuracy 0.33, F1 0.00 — chance level, showing CLIP similarity alone cannot handle idiomatic meaning.

2.5 Model 1 — Similarity MLP

Three features derived from CLIP outputs: cosine similarity, image L2-norm, text L2-norm. Network: Linear(3→16) + ReLU + Linear(16→1). BCEWithLogitsLoss pos_weight=2.0, Adam lr=0.001, 20 epochs. Group-wise sigmoid ranking at inference. Test: Accuracy 0.63, F1 0.44.

2.6 Model 2 — Full Embedding MLP (231 samples)

Full 1024-d CLIP vector as input. Network: Linear(1024→128) + ReLU + Linear(128→1). Same loss, optimiser, and group prediction as Model 1; 20 epochs on 231 samples. Test: Accuracy 0.93, F1 0.89.

2.7 Model 3 — Full Embedding MLP (690 samples)

Same as Model 2, trained on 690 augmented samples for 15 epochs, pos_weight=2.01. Test: Accuracy 0.93, F1 0.89 — matching Model 2. CLIP encodes meaning rather than exact wording, so paraphrases map to nearly identical embeddings; more training steps, same feature space.

2.8 Results

Approach	Accuracy	Precision	Recall	F1
Zero-shot — CLIP similarity	0.33	0.00	0.00	0.00
Model 1 — Similarity MLP	0.63	0.44	0.44	0.44
Model 2 — Full	0.93	0.89	0.89	0.89

Embedding MLP (231)					
Model 3 — Full Embedding MLP (690)	0.93	0.89	0.89	0.89	

The main gain came from moving from three summary features (Model 1, Acc 0.63) to the full 1024-d embedding (Model 2, Acc 0.93). Augmentation in Model 3 held that level without regression.

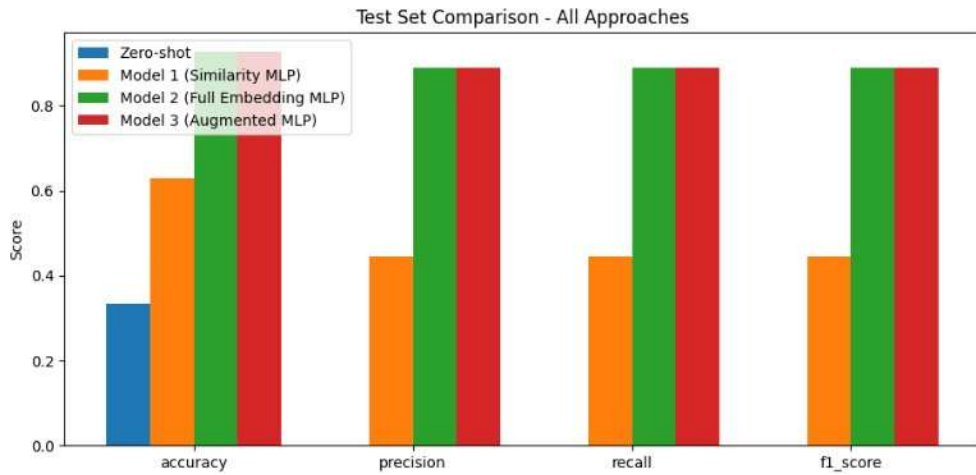


Fig. 6 — Test set comparison across all four approaches: accuracy, precision, recall, F1.

References

- [1] X. Wu et al., "A large-scale benchmark for food image segmentation," in Proc. ACM Int. Conf. Multimedia (MM), 2021. [Online]. Available: <https://dl.acm.org/doi/pdf/10.1145/3474085.3475201>
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Proc. MICCAI, 2015.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE CVPR, 2016.
- [4] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," J. Mach. Learn. Res., vol. 21, no. 140, pp. 1–67, 2020.
- [5] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. ICML, 2021.